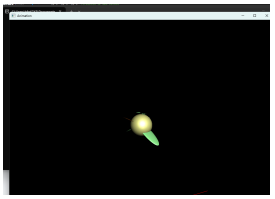
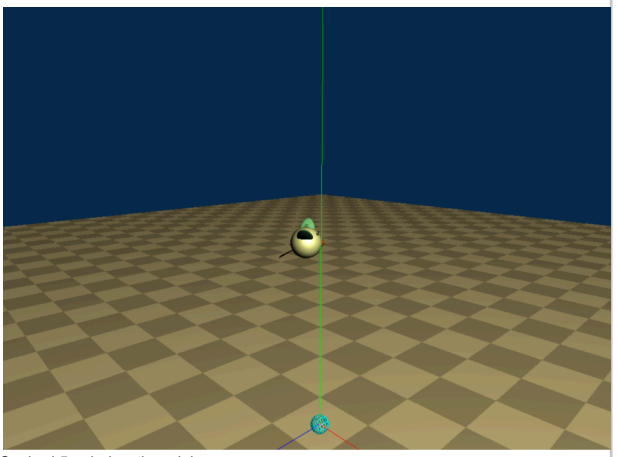
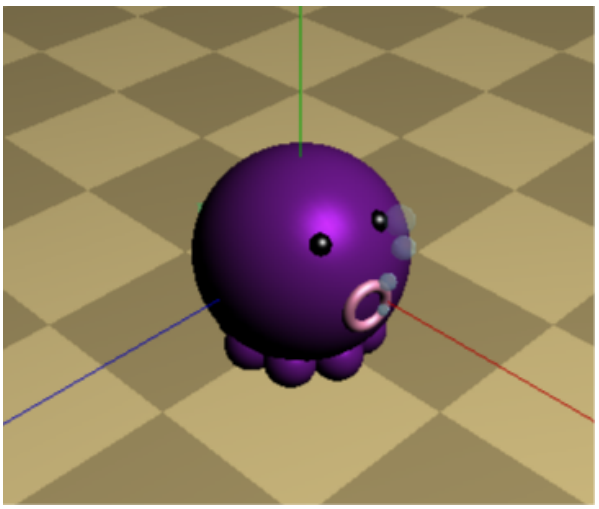
08/09/2026

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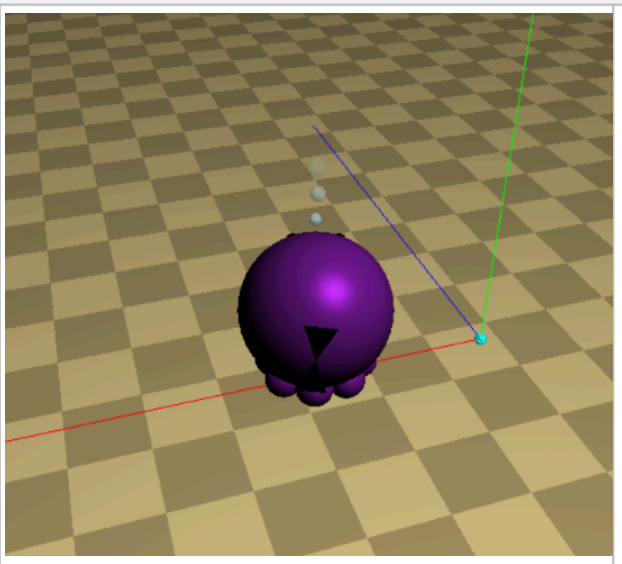
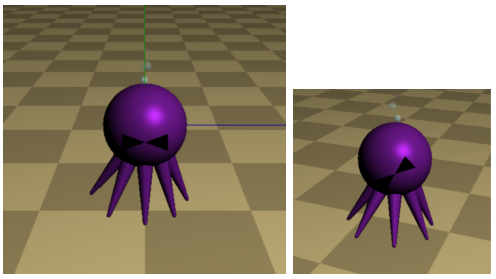
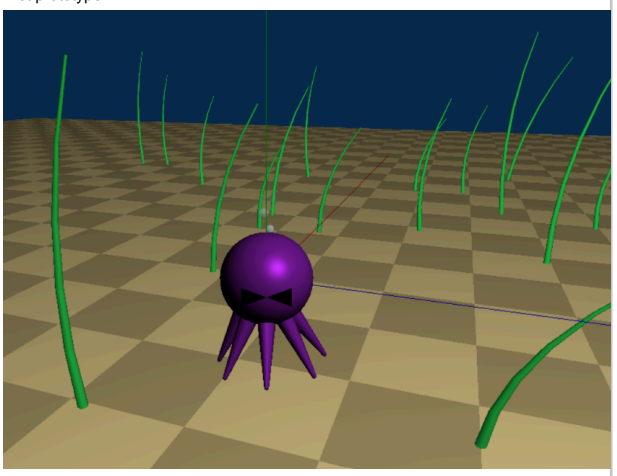
Time Spent	Date	To do	Work done	Detail	Bugs
2 hrs	5 Sept 20… 2 - 4PM	☑ *Animation Controller* Lecture	- Watch Module 14 - Create Project - Set up Basic Scene Settings	Setting up the project: - Animationcoller.c - Camera set up - Library referenced - Main - - **Double buffering:** glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH) in main, with glutSwapBuffers() at the end of display(). - **Depth buffering:** GLUT_DEPTH is requested, glEnable(GL_DEPTH_TEST) is in init(), and glClear(... | GL_DEPTH_BUFFER_BIT) runs every frame. - **Perspective and reshape:** reshape() calls glViewport and gluPerspective with the aspect ratio taken from the window size. 1.0GL unit = 1 meter.	- Freeglut.h not found Created the project not following the correct order. Fixed by restarting project set up.
1hr	7 Sept 20… 10 - 11AM	☑ ~~Body~~	- Bird.c Template - Set scale	I started by using the Bird.c script as a template  The bird moves around with arrow keys and space to stop/go. Change the scale for the bird to be 0.5 instead of 1.0. Instead of changing all the values I added: glScaled(0.5, 0.5, 0.5); before drawBird(); in Display(). **Body:** 1 m across **Wingspan:** about 2.5 m. **Tail:** 1 m. **Eyes:** 5 cm. **Propeller blades:** 0.25 m. In `reshape()`, change `1.0` to `0.1`. A 1.0 m near plane is as big as the bird's whole body, so it would clip the bird if the camera got close.	
1hrs	8 Sept 20… 2PM - 3PM	☑ ~~Seabed~~	- Create Grid - Set ocean color.	The seabed is a grid of squares so I can scale appropriately..  Seabed 5 m below the origin. The grid is `SEABED_SQUARES_PER_SIDE` squares of `SEABED_SQUARE_SIZE` metres each. The grid is subdivided into many small squares because OpenGL lights each vertex, so many small squares light more smoothly than a few big ones.	

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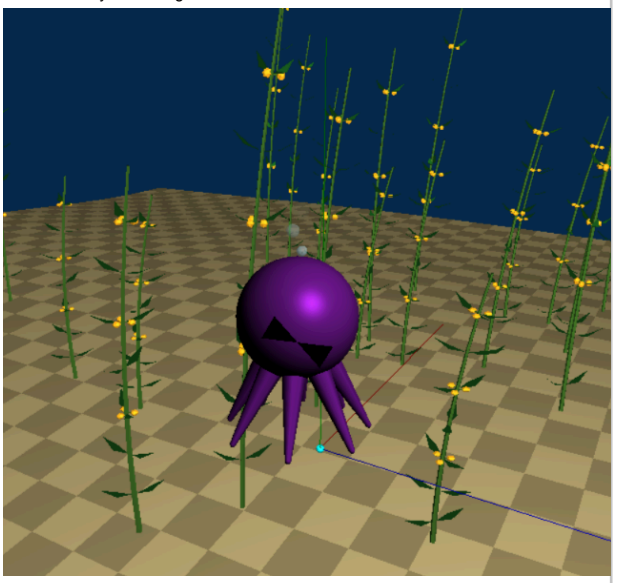
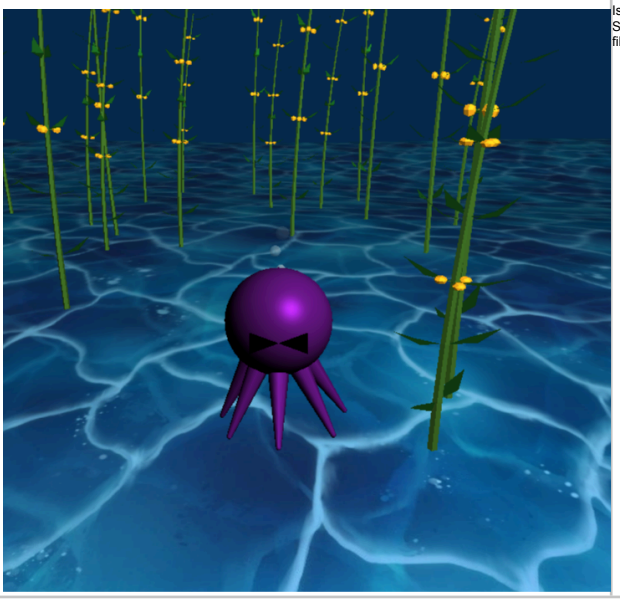
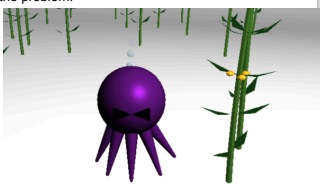
1hrs	15 Sept 2…
	2PM - 4PM
		☑ ~~Octopus~~	- Change Bird parts to octopus parts.
			- 8 Legs
			- Torus mouth
			- Propeller at back.
			
			- Animated bubbles
				
				
				Animated bubbles:
				 
				Toggle using b
				
				Changing the propeller spin to automatically toggle when octopus moves instead of "r" key.
				
				`if (move)`
				`propeller_angle = fmodf(propeller_angle + PROPELLER_DEG_PER_SEC * FRAME_TIME_SEC, 360.0f);`	

2hr	22 Sept 2… 2PM - 4PM	☐ Octopus tracking & control	**Octopus control (Can move in different directions)** Tracking Camera (camera always behind the octopus)	Make your octopus swim: Surge -- move forwards Sway -- move from side to side (left and right—in game terms, "strafe") Heave -- move directly up and down Yaw -- turn its heading (left or right) Used example code from 21a - Keyboard Motion Control.pdf to write. Octopus uses `float objectLocation[3] = { 0, 0, 0 }; // x, y (up), z`	

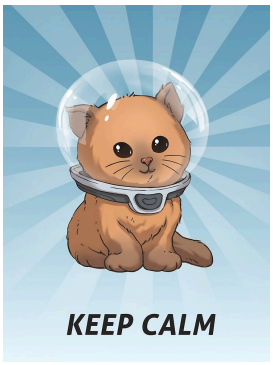
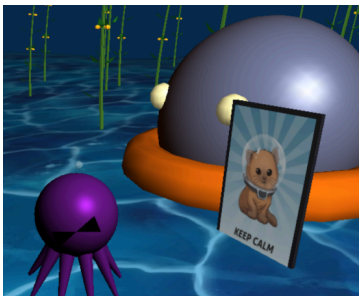
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				 Movement now waits for a `spinLevel` value (0 = stopped, 1 = full speed). It ramps up while a movement key is held and winds down when you let go. The propeller and legs spin at a speed scaled by `spinLevel`, and the octopus can only translate once `spinLevel` reaches a threshold. **Delay before moving:** with a 0.6 s spin-up and a 0.5 threshold, the octopus starts moving about 0.3 s after you press a key. It stops moving the moment you release, while the spin winds down. yaw works with the propeller and legs off.	
1hr	24 Sept 2… 10AM - 11AM		Seabed movement	Clamped the octopus's height in `think()` so its lowest point can't go below the seabed.	
6hr	29 Sept 2… 10AM - 4PM		Animated tentacles	Inspired by previous students' work I decided to change the octopus legs to look like tentacles and have a constant moving animation to mimic staying afloat in water. I kept the spinning legs logic when moving.  The spin is gradual speeding up and gradually coming to a stop same as propeller.	
4hr	9 Oct 2026 9AM - 1PM		Display lists - Seaweed - Seabed(sand dunes)	I used the displayLists example code to make my own seaweed. First prototype:  Animated to sway. I didn't like how it looked like grass so I took inspiration from my favorite underwater game "Subnautica" to create seaweed that looked similar to creepvines:	

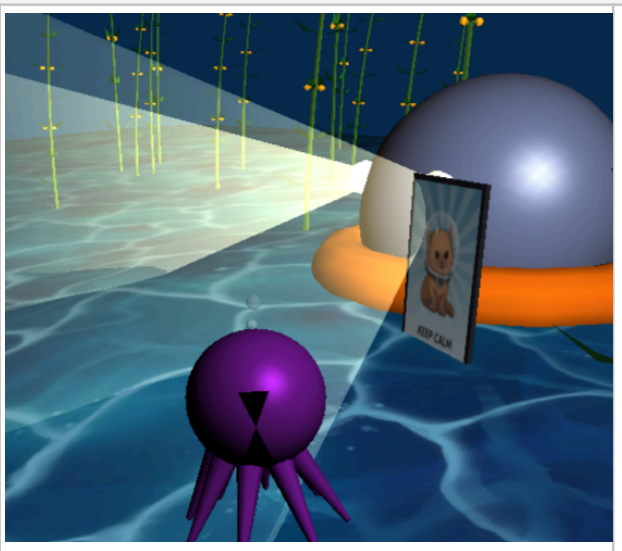
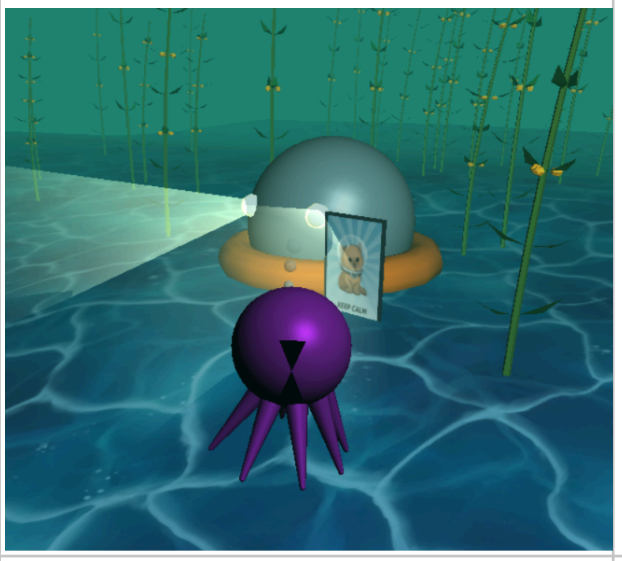
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				 So this was my final design:  Also animated by swaying.	
7hr	13 Oct 20… 10AM - 5PM	Texture Mapping PPM	Seabed terrain	I used the wrapTexture example to wrap in the *s* and *t* co-ordinates using GL_REPEAT. I decided on this for my seabed because the texture tiles across the whole terrain. This is the image I've decided with:  	Ppm texture failing to load! I added some print to console debugs to try figure out the problem.  Issue was image was actually a JPEG file. So I've installed irfanView to save the image as a ppm file.

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4hr	14 Oct 20… 3PM - 7PM	BPM texture mapping - poster		This poster is also from subnautica, I'm adding the texture to a flat rectangle to be a poster. I also am creating sunken lifepod for decorational objects. I had the idea for the animated spotlights to be flickering headlights on the lifepod.  This is my BPM texture.	
1hr	18 Oct 20… 7pm - 8pm	Materials		I deleted GL_COLOR_MATERIAL to use materials, my objects in my scene already use materials.	
8hr	19 Oct 20… 11am - 7pm	Animated spotlights		 The lights are pretty scary so I made them bigger and trying to make them look more like headlights.  The cone of the spot lights aren't visible so I change my global lighting settings to a darker setting. After no change I realized OpenGL spotlights only light surfaces, so the beam itself is invisible. To make it visible I drew a translucent cone shape from each headlight and fade it out along its length. It uses additive blending so it looks like light in water, and it switches off with the flicker.	

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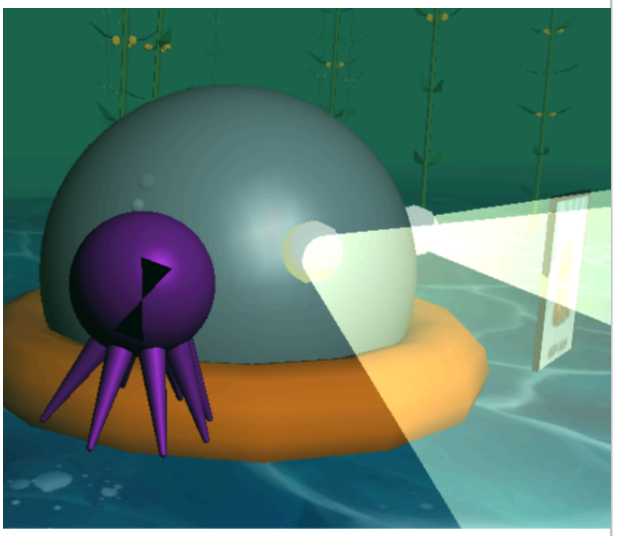
					
5hr	23 Oct 20… 8am - 1am	Fog		Making the fog there's a lot of blue in my scene but that doesn't really match the creep vines I have. In Subnautica the biome switches to very green atmosphere. I added this through the fog making it dense, murky, and green. 	
		Extra Features		1. Creep Vine Seaweed Low-poly stalks with leaves and pods, drawn from a display list, swaying with a looping sine phase.  2. Bubbles	



3. Keep Calm Poster



4. Crashed LifePod



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Reflection

Design choices. I built a low-poly plum octopus from quadrics, working in metres so the camera, speeds and sizes were easy to reason about. One function, seabedHeight(), drives the terrain, seaweed, octopus floor and lifepod, so nothing floats or sinks when the dunes change. The creepvine seaweed and the sphere lifepod with its orange ring give the scene its identity.

Adjustments. I replaced the flat checkerboard seabed with textured dunes and flattened the area around the pod. I made the fog follow the camera depth and position, with a slight green tint. I also added pod collision and a seaweed exclusion zone.

Errors and lessons. My seabed texture failed because the file was a renamed JPEG, which I only found by printing its first two bytes. The seaweed snapped at the end of each cycle because I scaled a wrapped phase by a non-whole number. A stray drawTentacle() call without push/pop shifted every object drawn after it. Lights and fog also had to be set after gluLookAt.

Next time. I would test each loader straight away, plan the coordinate layout on paper first, and save a working copy before every big change.

Shortcomings. The light cones are drawn shapes that sand and the pod don’t block. Collision is sphere-only, so the octopus can pass through the poster or slip past the ring. Each seaweed strand sways rigidly, and texture paths are hard-coded.